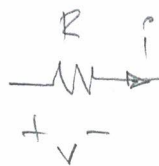


4.31)



$$P = Vi = Ri^2$$

BESTÄM EFFEKTEN SOM DISSIPERAS I EN RESISTOR

Lösning:

$$P(t) = Ri(t)^2 = 1 \cdot 10^2 \cdot \sin^4 t \quad \text{och} \quad P_{\text{rms}} = Ri_{\text{rms}}^2$$

RMS VÄRDEN AV i BLIR

$$\begin{aligned} i_{\text{rms}}^2 &= 100 \cdot \frac{1}{T} \int_0^T \sin^4(t) dt = \\ &= 100 \cdot \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} (1 - \sin(2t)) \frac{1}{2} (1 - \sin(2t)) dt = \\ &= 100 \cdot \frac{1}{8\pi} \int_0^{2\pi} [1 - 2\sin(2t) + \sin^2(2t)] dt = \\ &= 100 \cdot \frac{1}{8\pi} \left\{ [2\pi - 0] + \frac{1}{2} \int_0^{2\pi} (1 - \sin(4t)) dt \right\} = \\ &= 100 \cdot \frac{1}{8\pi} \{ 2\pi + \pi \} = 100 \cdot \frac{3}{8} \approx \end{aligned}$$

$$\Rightarrow i_{\text{rms}} = 10 \sqrt{\frac{3}{8}} \approx \underline{6.12} \text{ A}$$

$$\text{och} \quad P_{\text{rms}} = Ri_{\text{rms}}^2 = 1 \cdot (6.12)^2 \approx \underline{37.5 \text{ W}}$$

SVAR: $P_{\text{rms}} = 37.5 \text{ W}$